Dr. Monolina Sarkar

monolinasarkar1@gmail.com | www.linkedin.com/in/monolinasarkar
ORCID ID: orcid.org/0000-0002-8172-1111 | Web of Science ID: GRY-0666-2022

Research Profile

Science education researcher and Life Science educator with over 10 years of experience teaching undergraduate and postgraduate students in India. Research interests include student engagement, motivational climate, first-generation learners, and women in inclusive teaching. Combines a PhD in Biochemistry with formal training in Education Technology and active involvement in higher education research.

.

Education

- **M.A. Education** 2024
  School of Education, Indira Gandhi National Open University
  Specialization: Education Technology
- **Ph.D. Biochemistry** 2020
  Department of Biochemistry, University of Calcutta
- **M.Sc. Botany** 2011
  Department of Botany, University of Calcutta
  Specialization: Cell Biology and Ecotoxicology
- **B.Sc. Botany (Honors)** 2009
  Lady Brabourne College, Department of Botany, University of Calcutta
  General Electives: Chemistry, Zoology

EXPERIENCE

- **State-Aided College Teacher-(Category-I)** 01/2020 – Present
  - Sammilani Mahavidyalaya (affiliated with University of Calcutta)
    B.Sc. Botany Honours
    B.Sc. Environmental Studies
    - Teach undergraduate biology courses using laboratory-based and active-learning approaches to enhance student engagement and conceptual understanding.
    - Design and implement course curricula, instructional materials, practical sessions, assessments, and learning activities aligned with undergraduate science education objectives.
    - Continuously evaluate student learning and provide formative feedback to improve academic performance and critical thinking.
    - Coordinate course delivery and collaborate with faculty to ensure effective implementation of academic programs.
    - Developed a strong interest in evidence-based teaching practices to improve student engagement and learning in undergraduate biology classrooms.

- **Visiting Lecturer**
  - West Bengal State University 03/2022 – 04/2025
    M.Sc. Biochemistry Department
  - Diamond Harbour Women's University 01/2019 – Present
    M.Sc. Botany Department
  - Brahmananda Keshab Chandra College 08/2017 – 1/2024
    West Bengal State University
    M.Sc. Botany Department
  - Sammilani Mahavidyalaya 08/2018 – 12/2019
    B.Sc. Botany Department
  - JIS University 01/2017 – 07/2017
    M.Sc., B.Sc. Biotechnology Department

Supervised Dissertation Topics

- Supervised MSc Cell Biology students in literature-based research projects, providing guidance in scientific literature evaluation, research question formulation, scientific writing, interpretation of biological concepts, and research presentation.
- The supervision experience provided valuable insights into student learning approaches, conceptual challenges, and the development of scientific reasoning among postgraduate learners, contributing to refinement of classroom teaching strategies.

Theme: Cell Signaling and Molecular Pathway Regulation

- Mukherjee, S. (2024). *Effect of eugenol, nimbin, and berberine on glucose metabolism.*
- Majhi, S. (2024). *Therapeutic effect of resveratrol on signaling pathways during alcohol intoxication.*
- Giri, S. (2024). *Cell signaling pathways regulating glycogen storage.*
- Saha, A. (2023). *Therapeutic potential of resveratrol in Parkinson's disease.*
- Ghosal, S. (2023). *PI3K/AKT signal pathway and the role of curcumin in the prevention and treatment of Alzheimer's disease.*

Theme: Plant–Microbe Interactions and Symbiosis

- Salui, S. (2022). *Ecological significance of bacteria-bryophyte symbiotic interaction in land plant evolution.*
- Mandal, S. (2022). *Diversity of bacteria-bryophyte symbiotic interactions.*

PUBLICATIONS

- Sarkar, Monolina. "Light-responsive biodegradation of wastewater pollutants: New developments and potential perspectives." *Journal of Hazardous Materials* Advances 10 (2023): 100281.
- Das, Debapriya Rajlakshmi, Monolina Sarkar, and Anindita Paul. "Tools and approaches for assessing stress-responsive secondary metabolites to design climate-smart crops." *In Biology and Biotechnology of Environmental Stress Tolerance in Plants,* pp. 305-370. Apple Academic Press, 2023.
- Sarkar, Monolina. "Effect of silver nanoparticles on nitrogen-cycling bacteria in constructed wetlands." *Nanotechnology for Environmental Engineering* 7, no. 2 (2022): 537-559.
- Sarkar, Monolina, Anindita Paul, and Debapriya Rajlakshmi Das. "Availing engineered and biosynthesized metal-based nanoparticles to alleviate abiotic stress." *Response of field crops to abiotic stress* (2022): 15-33.

- Guha, Sohini, Firoz Molla, Monolina Sarkar, Fernando Ibañez, Adriana Fabra, and Maitrayee DasGupta. "Nod factor-independent 'crack-entry'symbiosis in dalbergoid legume *Arachis hypogaea*." *Environmental Microbiology* 24, no. 6 (2022): 2732-2746.
- Guha, Sohini, Monolina Sarkar, Pritha Ganguly, Md Raihan Uddin, Sukhendu Mandal, and Maitrayee DasGupta. "Segregation of nod-containing and nod-deficient bradyrhizobia as endosymbionts of *Arachis hypogaea* and as endophytes of *Oryza sativa* in intercropped fields of Bengal Basin, India." *Environmental Microbiology* 18, no. 8 (2016): 2575-2590.

PEER REVIEWER EXPERIENCE

- Environmental Science and Pollution Research (Springer Nature)
- Journal of environmental management. (Elsevier)
- American Chemical Society Reviewer Lab, Reviewer Certification

CONFERENCE PRESENTATIONS

- **Science Talk'23**, Portland 2023
  Talk: Promoting science communication in Higher Education
  The Association of Science Communicators
  Theme: Giving Science a Voice
- **Biodiversity - Stewardship for Vital Resources** 2022
  Poster: Engineering the plant microbiome to combat abiotic stress in plants
  Dresden Nexus Conference 2022, Germany
  Institute for the Integrated Management of Material Fluxes and of Resources, Technische Universität Dresden, and Leibniz Institute of Ecological Urban and Regional Development

AFFILIATIONS

- Member, UNESCO- Inclusive Policy Lab
- Member, MERLOT: Multimedia Educational Resource for Learning and Online Teaching
- Member and Web Writer, American Society of Microbiology

PROFESSIONAL DEVELOPMENT 2022 – Present

- **Johns Hopkins Teaching Institute Workshop| 2026**
  - Developed course framework using active learning principles
  - Strengthened capability in learner-centered pedagogy and curriculum architecture

- **Accessible Science Communication**
  - **American Society for Microbiology | 2023 – Present**

    The Alchemy of Fine Chocolate: How Microbes Shape Flavor | May 12, 2026
    Microbial Succession and Post-Fire Recovery | August 6, 2025
    Predatory Bacterium Feasts on Soil Microbiome | July 17, 2023
  - **CSIR – National Institute of Science Communication and Policy Research | 2015**

    How Plants Turn Their Wounds into Wisdom
    Science Reporter, 52(03), 48–49

- **Data Analysis & Research Training**
  - Hands-on training in Next-Generation Sequencing (NGS)
    Center for Climate and Environmental Studies
    Indian Institute of Science Education and Research (IISER), Kolkata
    International Society for Microbial Ecology (ISME)
  - Biostatistics: A User's Perspective
    Indian Institute of Science Education and Research (IISER), Pune

- **Learning Practices & Pedagogical Training**
  - Blended Learning Practice
    Commonwealth of Learning, Canada | Athabasca University, Canada
  - Understanding Open Educational Resources
    Commonwealth of Learning, Canada
  - Inclusive STEM Teaching Project, The Inclusive STEM Teaching Project Team
    Boston University | Northwestern University
  - Advancing Learning Through Evidence-Based Teaching
    Center for the Integration of Research, Teaching, and Learning (CIRTL)
    Wisconsin Center for Education Research, University of Wisconsin–Madison

OUTREACH PROJECTS (2022-2024)

- **Instructional Designer**
  CrowdDoing, San Francisco, CA
  - Assess learning needs via surveys and in alliance with Learning and Development lead.
  - Create learner profiles to develop relevant and engaging learning experiences.
  - Design, develop, implement, and evaluate learning solutions.
  - Research and develop multimedia learning resources.
  - Review quality assurance to identify and address improvements in learning solutions.
- **Curriculum Developer & Subject Matter Expert**
  eVidyalok, India
  - Contributed biology instructional content for rural students in West Bengal by developing a lesson aligned with the organization's curriculum framework.

DOCTORAL RESEARCH EXPERIENCE

- **Visiting Researcher Fellowship** 10/2015-12/2015
  Indo-French Centre for the Promotion of Advanced Research (IFCPAR)
  Campus International de Baillarguet, Montpellier
- **Senior Research Fellowship** 01/2014-12/2016
  Council of Scientific and Industrial Research, Ministry of Science and Technology, India
  Area of Research: Life Science
- **Junior Research Fellowship** 01/2012-12/2013
  Council of Scientific and Industrial Research, Ministry of Science and Technology, India
  Area of Research: Life Science
- **CSIR-National Eligibility Test (NET)** 06/2011
  Eligibility test for Junior Research Fellowship (JRF) and Lectureship/Assistant Professor
  Council of Scientific & Industrial Research and University Grants Commission, India
  Subject: Life Science, All India Rank-42

- **Graduate Aptitude Test in Engineering (GATE)** 03/2011
  Conducted by Indian Institute of Technology, Madras
  On behalf of the National Coordination Board, Graduate Aptitude Test in Engineering
  Department of Higher Education, Ministry of Human Resource Development, India
  Subject: Life Sciences, All India Rank-608